

Hadoop Installation and Configuration Guide

STEP 1 — Update Ubuntu

```
sudo apt update
```

STEP 2 — Install Java 8

```
sudo apt install openjdk-8-jdk -y
```

STEP 3 — Check Java Version

```
java -version
```

Select Java 8 if multiple versions exist:

```
sudo update-alternatives --config java
```

Choose the number corresponding to Java 8 and press Enter.

STEP 4 — Go to Home Directory

```
cd ~
```

STEP 5 — Download Hadoop

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

STEP 6 — Extract Hadoop

```
tar -xvzf hadoop-3.3.6.tar.gz
```

STEP 7 — Move Hadoop Folder

```
sudo mv hadoop-3.3.6 /usr/local/hadoop
```

STEP 8 — Configure Environment Variables

```
gedit ~/.bashrc
```

Add the following lines at the end of the file:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/usr/local/hadoop
export PATH=$PATH:$HADOOP_HOME/bin
export PATH=$PATH:$HADOOP_HOME/sbin
```

Save the file and run:

```
source ~/.bashrc
```

STEP 9 — Check Hadoop Version

```
hadoop version
```

STEP 10 — Configure Hadoop

```
cd /usr/local/hadoop/etc/hadoop
```

Edit core-site.xml

```
sudo gedit /usr/local/hadoop/etc/hadoop/core-site.xml
```

Add inside <configuration>:

```
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://localhost:9000</value>
</property>
```

Edit hdfs-site.xml

```
sudo gedit /usr/local/hadoop/etc/hadoop/hdfs-site.xml
```

Add:

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

STEP 11 — Format NameNode

```
hdfs namenode -format
```

STEP 12 — Install and Enable SSH

```
sudo apt install openssh-server -y
sudo service ssh start
sudo systemctl enable ssh
sudo systemctl status ssh
```

Connect to localhost

```
ssh localhost
Type yes
```

Generate SSH Key

```
ssh-keygen -t rsa -P ""
```

Copy Key

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
chmod 600 ~/.ssh/authorized_keys
```

Test Again

```
ssh localhost
Type exit to close connection.
```

STEP 13 — Start Hadoop First Time

```
start-dfs.sh
start-yarn.sh
jps
```

If restarting later:

```
stop-dfs.sh
stop-yarn.sh
start-dfs.sh
start-yarn.sh
jps
```

STEP 14 — Fix JAVA_HOME for Hadoop

```
readlink -f $(which java)
```

Copy parent Java folder path and remove /bin/java from end.

Edit files:

```
sudo nano /usr/local/hadoop/etc/hadoop/hadoop-env.sh
sudo nano /usr/local/hadoop/etc/hadoop/hdfs-env.sh
sudo nano /usr/local/hadoop/etc/hadoop/yarn-env.sh
```

Search JAVA_HOME and set:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/jre
```

Save using:

Ctrl + O → Enter → Ctrl + X

Reload bash:

```
source ~/.bashrc
```

Restart Hadoop:

```
start-dfs.sh
```

```
start-yarn.sh
```

```
jps
```

Expected jps Output

NameNode

DataNode

SecondaryNameNode

ResourceManager

NodeManager

Jps